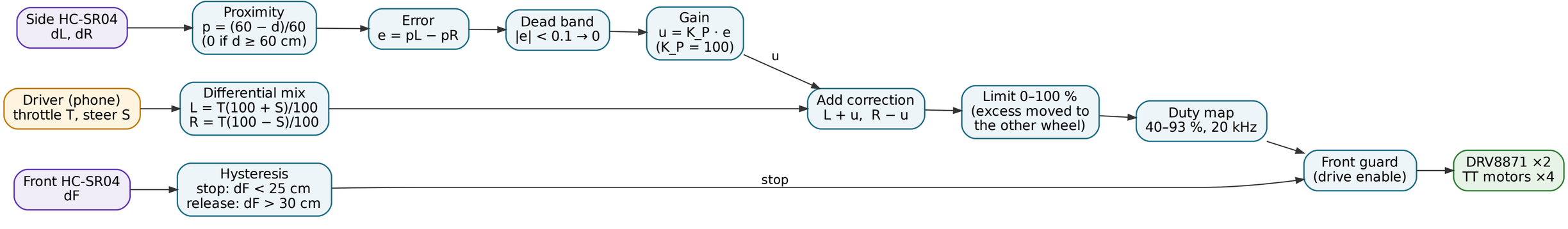
Side HC-SR04
dL, dR
Proximity
p = (60 − d)/60
(0 if d ≥ 60 cm)
Error
e = pL − pR
Dead band
|e| < 0.1 → 0
Gain
u = K_P · e
(K_P = 100)
u
Driver (phone)
throttle T, steer S
Differential mix
L = T(100 + S)/100
R = T(100 − S)/100
Add correction
L + u, R − u
Limit 0–100 %
(excess moved to
the other wheel)
Duty map
40–93 %, 20 kHz
Front HC-SR04
dF
Hysteresis
stop: dF < 25 cm
release: dF > 30 cm
stop
Front guard
(drive enable)
DRV8871 ×2
TT motors ×4